

EI - Limit Order Book prediction

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Overview



- Introduce the objectives
- Analysis and processing the data
- Features selection
- Presentation of two strategies and how to assess the strategies performance



Objectives of this study

- Study limit order books in the context of asynchronous order book data
- Design a cross-crypto strategy to predict the increase of ETH when influenced by other cryptocurrencies (Especially XBT)
- Backtest this strategy and assess the out-of-sample performances

Processing of the data :

Large volume of data how to process ?

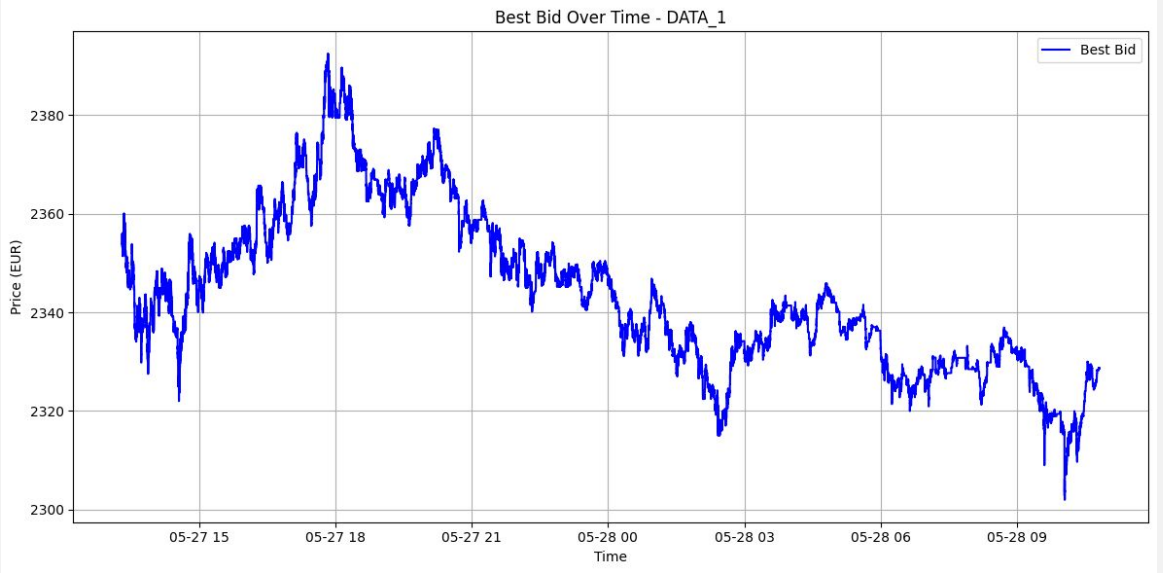


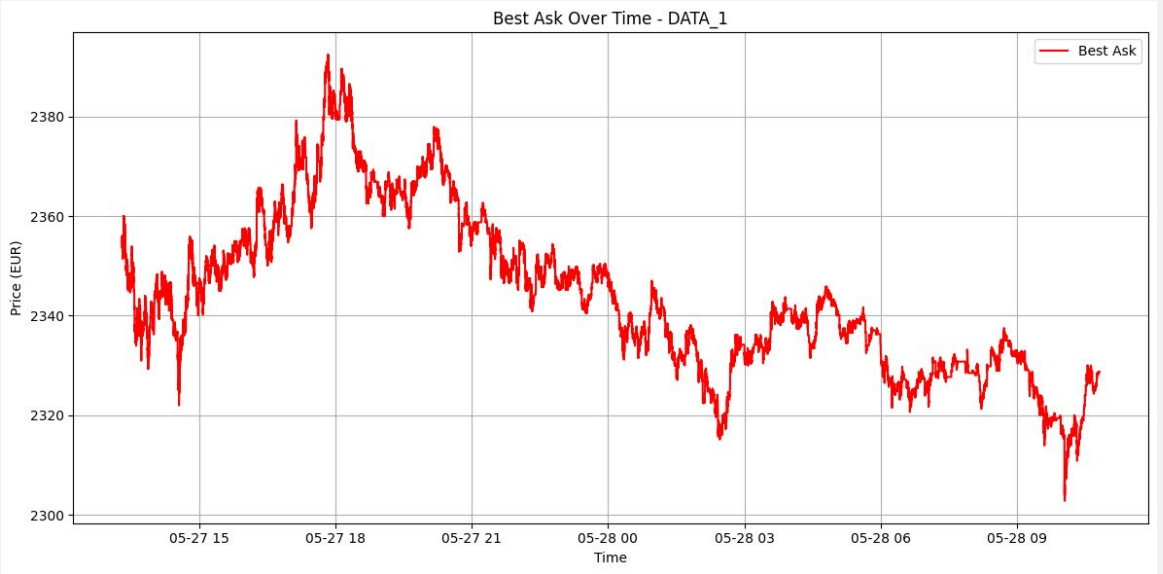
.csv → .parquet

Calculate only once by saving snapshots in .parquet

Preprocessing of the data

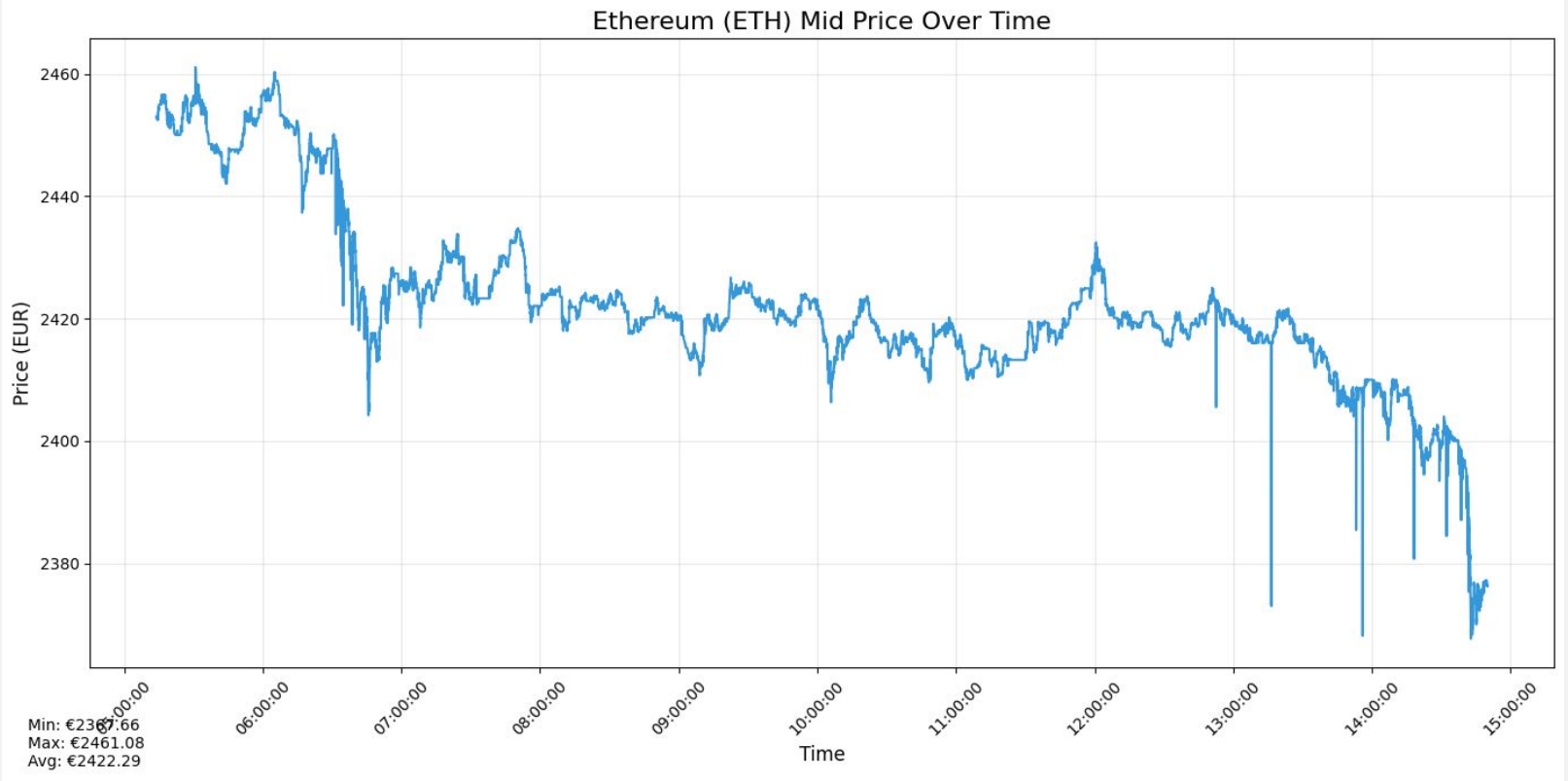
Extracting the best bid and best ask for each coin we are interested in





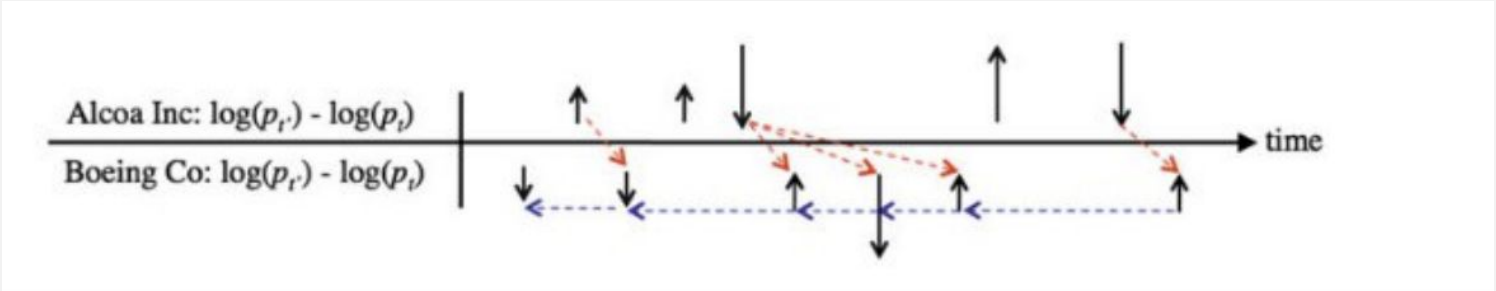
Which signal we want to study :

Mid price of ETH



Difficulties in cross-crypto analysis :

Alignment of XBT/ETH data

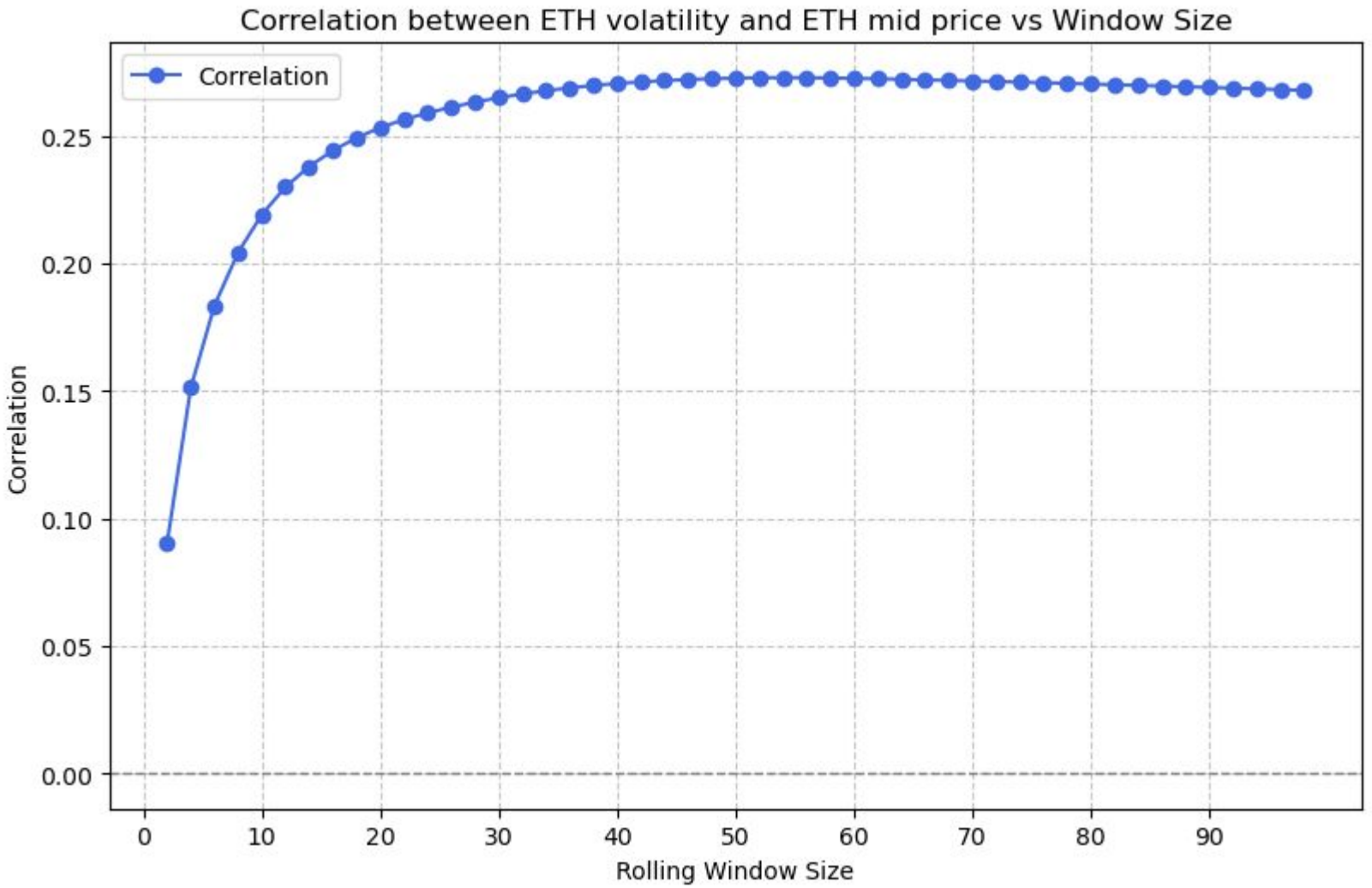


Features selection

What should we take into account in our model ?

1. Lead lag optimization : Find the optimal time lag between variables to improve predictive power
2. First Statistical inference : Correlation matrix to quickly identify potential linear relationships between variables
3. Formal Causality Testing : Statistically test if one variable causes another in a predictive sense.

Lead lag optimization

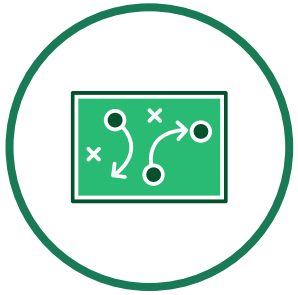


List of features :



Price based features

- Mid-prices for ETH and BTC
- ETH/BTC price ratio
- Price spreads (absolute and percentage)



Volume-based features

- Total volumes for ETH and BTC
- Volume imbalances (bid vs ask)



Time-series features

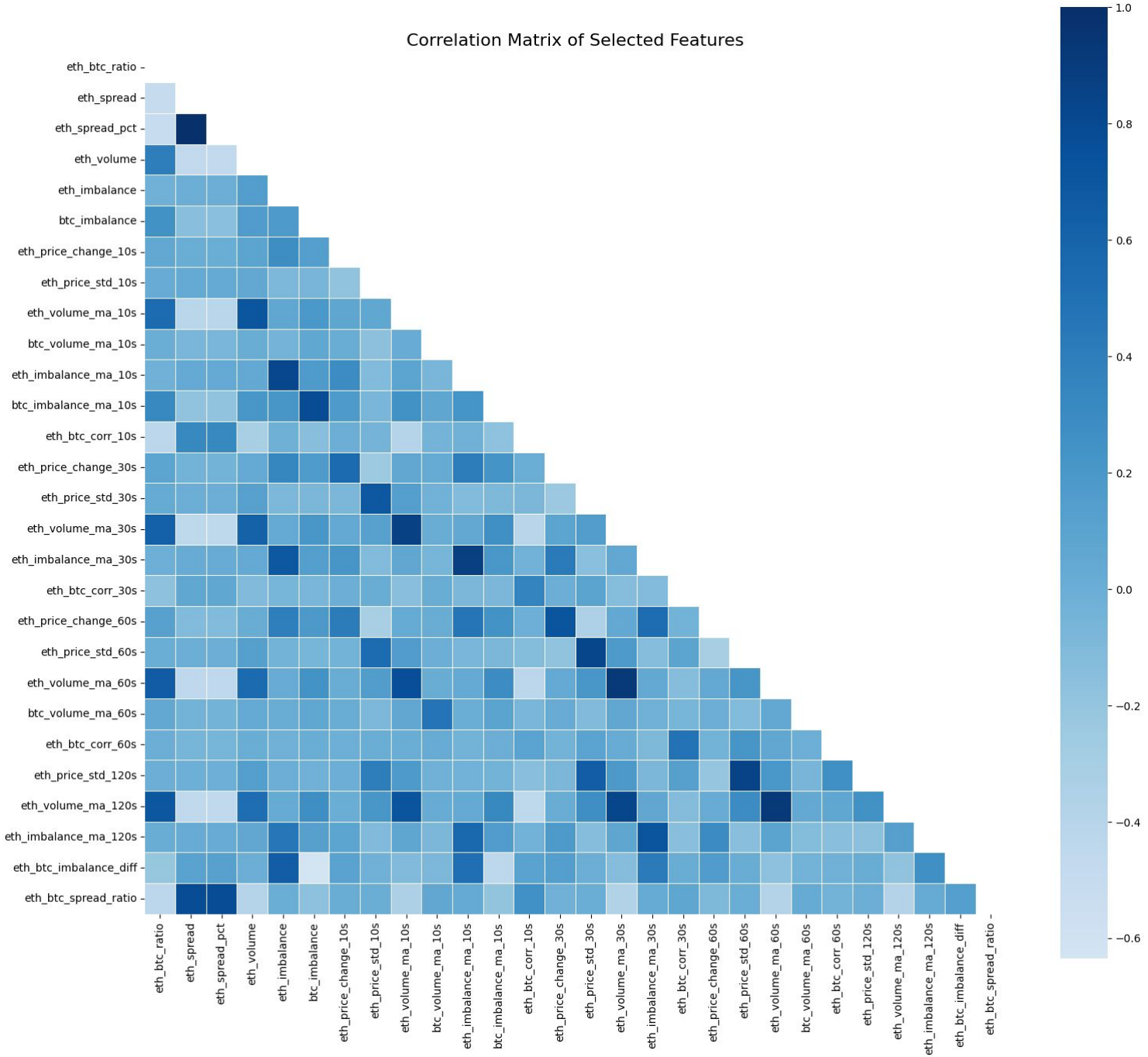
- Price percentage changes
- Price volatility
- Moving averages for prices
- Volume moving averages
- Imbalance moving averages



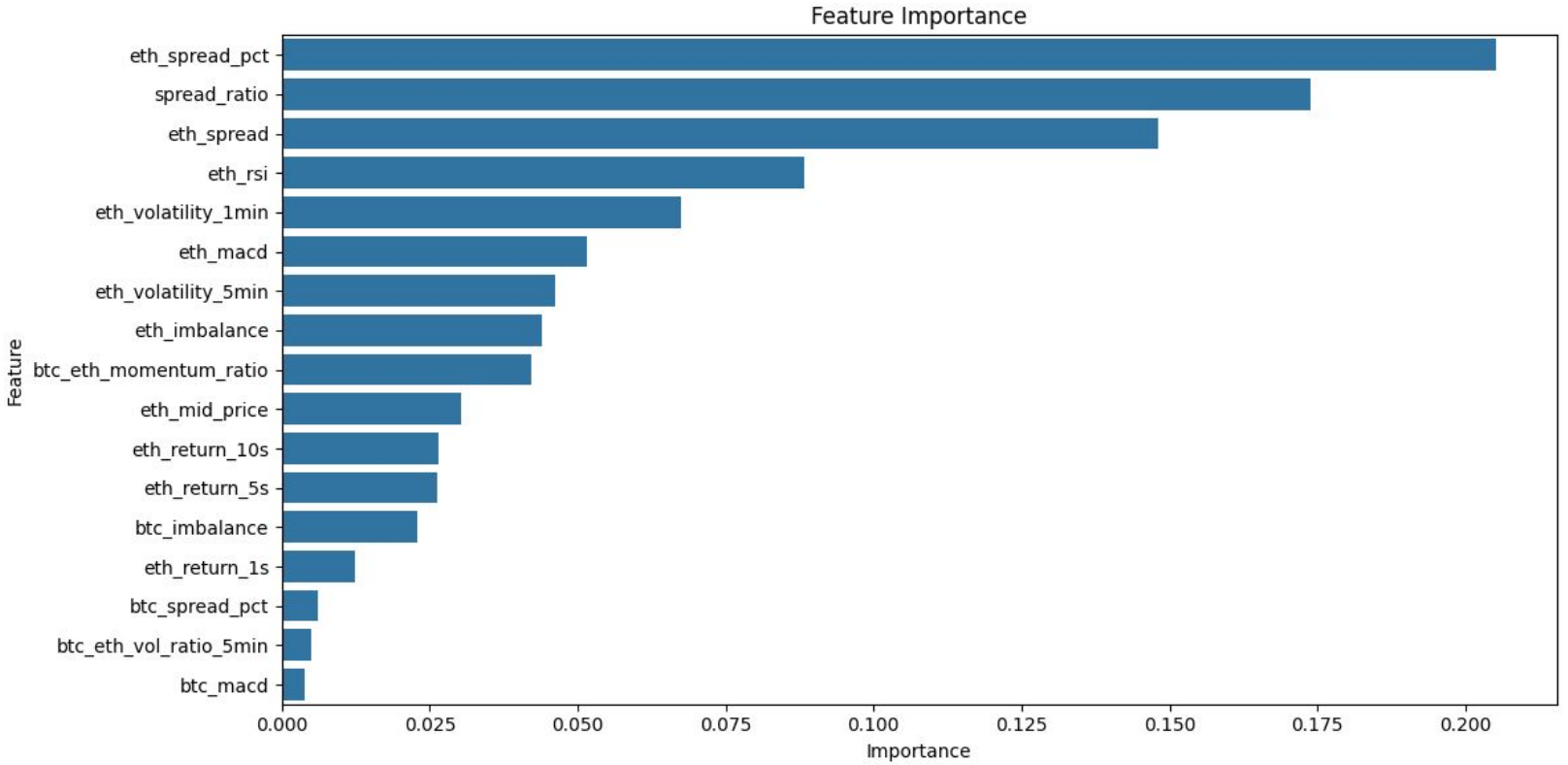
Cross-asset features

- ETH-BTC price correlations
- Imbalance differences
- Spread ratios

Correlation Matrix



Causality testing RandomForest :





Key Findings (1)

Short-term Volatility Dominates

The top predictive features are short-term price volatility indicators:

- eth_price_std_10s (10-second price standard deviation)
- eth_price_change_10s (10-second price percentage change)
- eth_spread and eth_spread_pct (absolute and relative bid-ask spreads



Key Findings (2)

Feature Selection :

Feature Selection: From the original 56 features, 28 were selected based on importance, reducing dimensionality while maintaining predictive power.



To uncover real cross-crypto predictive signals :

For each candidate pair (BTC feature, ETH -feature), and each lag (delay) : measure of the TE from BTC to ETH

We looped over many feature combinations and lags on the entire in-sample DATASET0

btc_feature	eth_feature	lag	te	p_value
return_10_symbol	imbalance_symbol	1	0.001482	0.0
return_10_symbol	imbalance_symbol	2	0.001480	0.0
return_20_symbol	return_20_symbol	2	0.001479	0.0
return_20_symbol	return_20_symbol	3	0.001467	0.0
return_20_symbol	return_20_symbol	1	0.001460	0.0
return_20_symbol	imbalance_symbol	2	0.001447	0.0
return_10_symbol	imbalance_symbol	3	0.001434	0.0
return_10_symbol	imbalance_symbol	4	0.001433	0.0
return_20_symbol	return_10_symbol	2	0.001424	0.0
return_20_symbol	return_20_symbol	4	0.001419	0.0
return_20_symbol	imbalance_symbol	1	0.001418	0.0
return_20_symbol	return_10_symbol	1	0.001418	0.0
return_10_symbol	return_20_symbol	2	0.001416	0.0
return_20_symbol	imbalance_symbol	3	0.001411	0.0
return_20_symbol	return_10_symbol	3	0.001409	0.0
return_10_symbol	return_20_symbol	1	0.001392	0.0

Causality Testing with transfer Entropy based analysis

Strategy



Transfer Entropy based Strategy :

Return_10_symbol :
midprice(t)-midprice-(t-10)

Cross-crypto lead-lag signal : BTC Return_10_Symbol and ETH imbalance

Question : Is it a noise or a rea signal ?



Realistic backtest :
OOS DATASET 1

Based on the TE result :

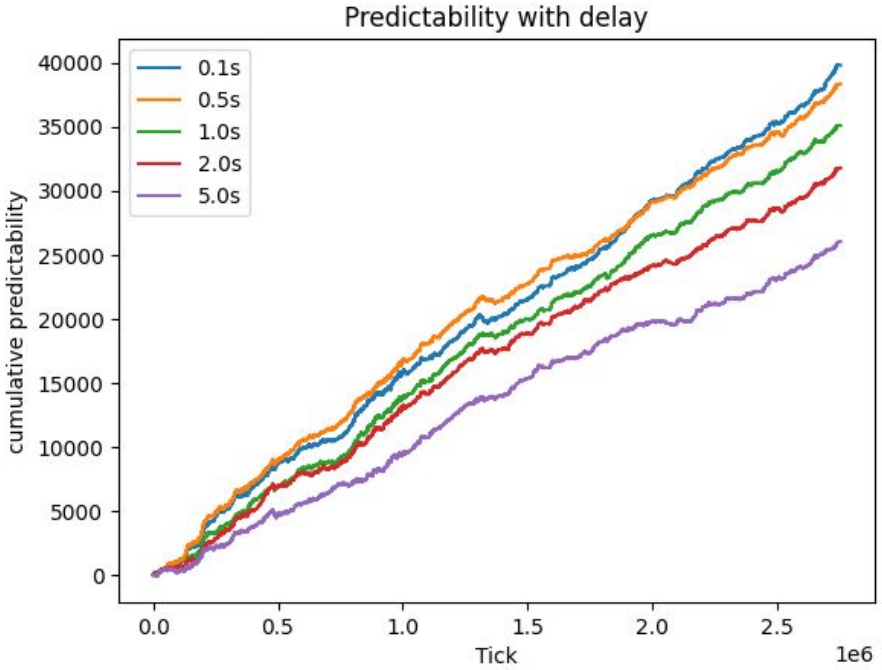
Simple actionable signal

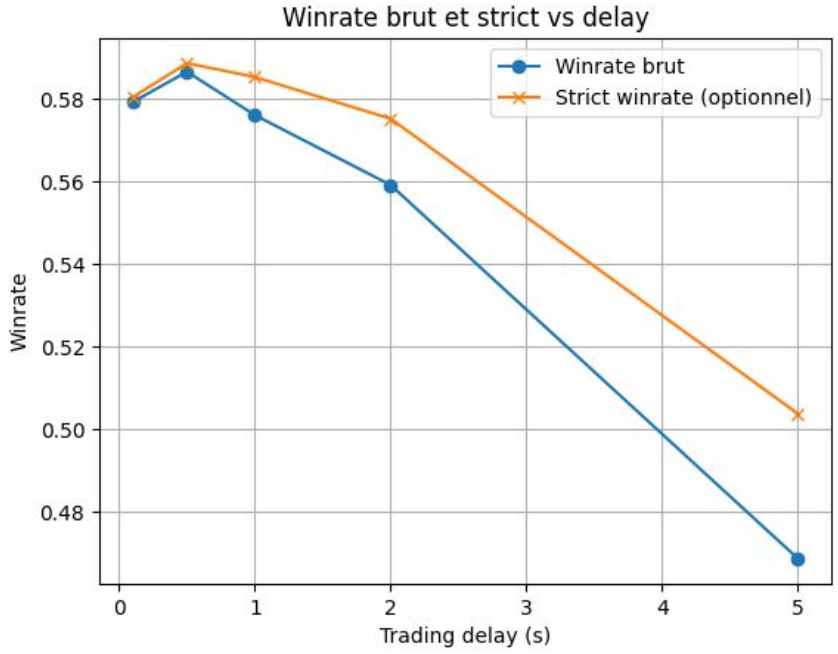
BTC UP -> Long ETH

BTC DOWN -> Short ETH

BTC FLAT -> No position

$$P = \sum_k a_k \quad \text{with} \quad a_k = \begin{cases} 1 & \text{if the prediction is correct} \\ -1 & \text{if the prediction is incorrect} \\ 0 & \text{if the prediction is indeterminate} \end{cases}$$





Impact of transaction-delay on performance :



Strategy :

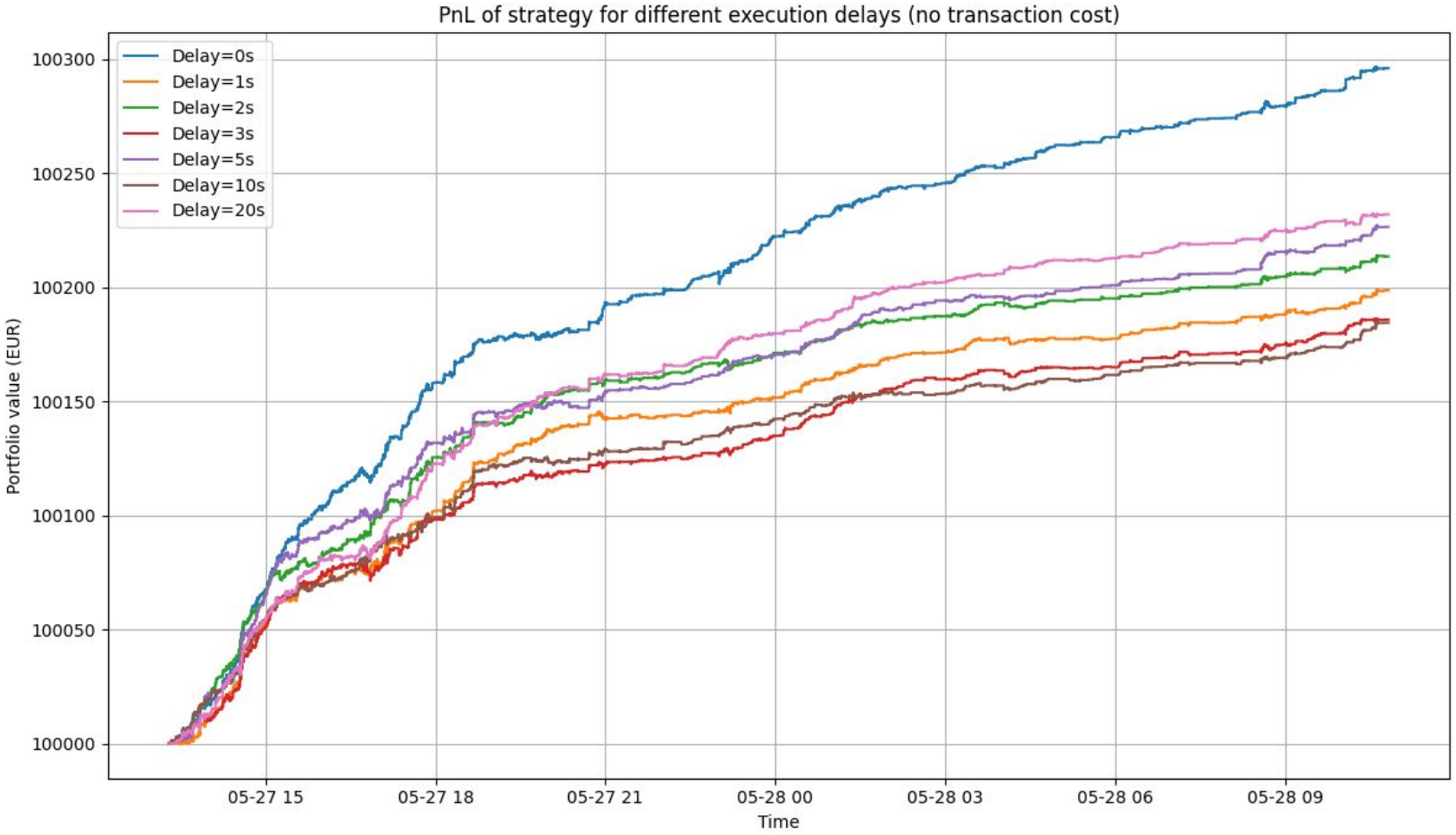
Trading Rule :

- Go LONG ETH when BTC UP
- Go SHORT ETH when BTC DOWN
- Stay FLAT when BTC FLAT
- EXIT/REVERSE the position when the signal changes

Portfolio Management :

- Fixed amount of Cash (100_000)
- Profits and losses : realized as the position is closed/reversed
- Portfolio value marked to market at every step
- The strategy does not compound or leverage positions each trade uses a fixed notional size

Without exchange fees :





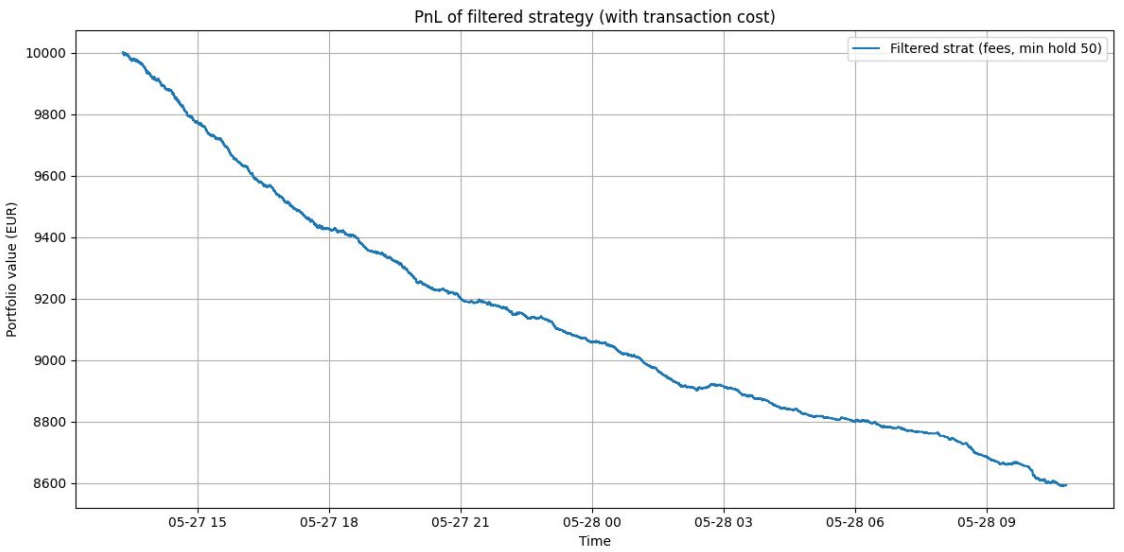
Including realistic transaction costs

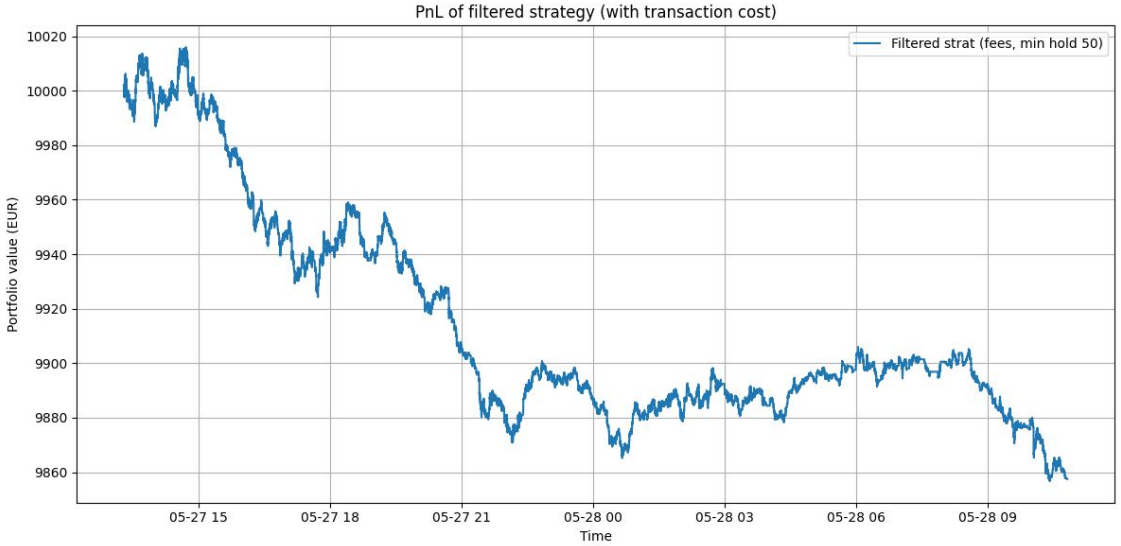
We evaluate the strategy using these assumptions:

- **Exchange fee**: 0.2% per trade
- **Slippage**: 0.1% per execution
- **Execution delay**: 1 second



Including realistic transaction costs

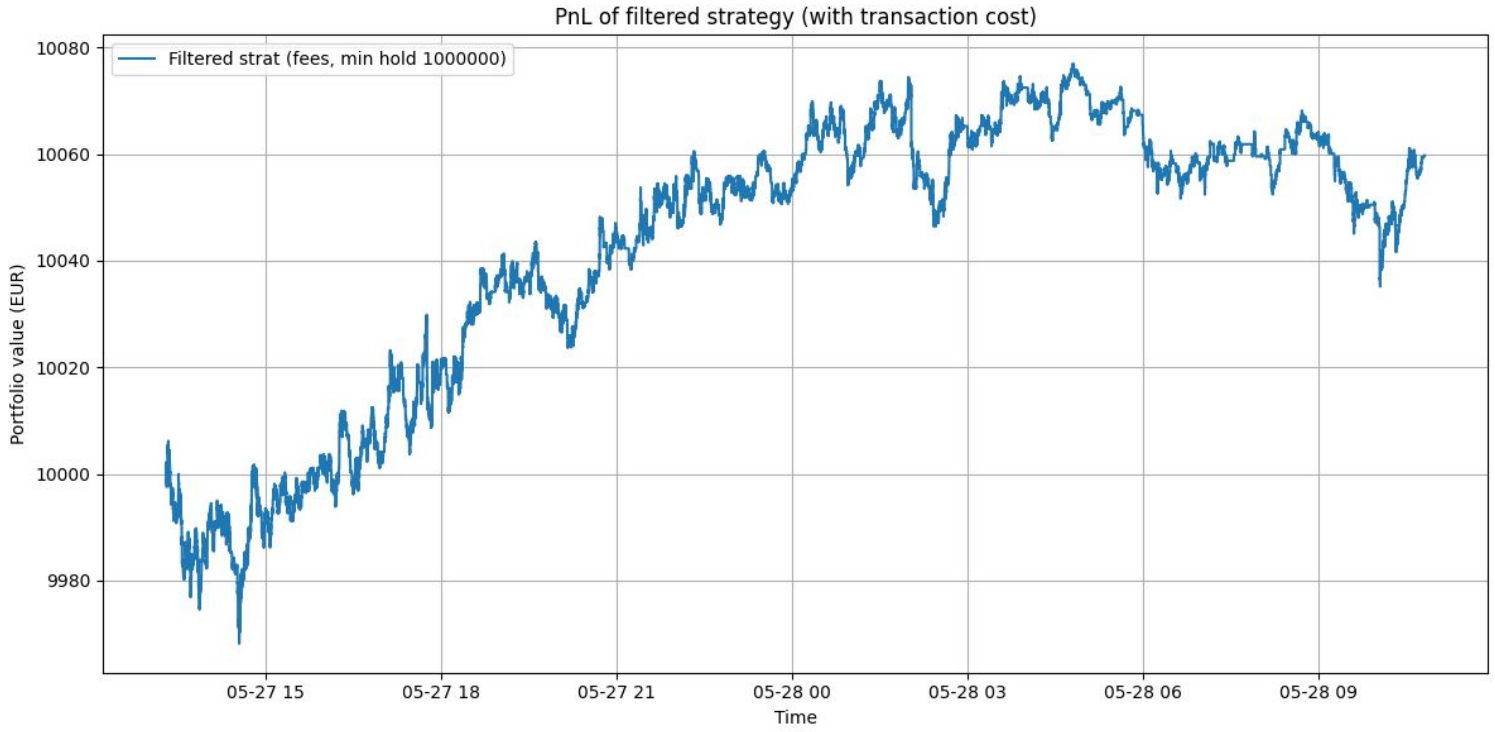




1 tick = 0,0005s



Including realistic transaction costs



1 millions tick hold = 500s : not relevant for HFT

Total Brut perf: 0,5973%
Total Backtest duration : 21,52h
Alpha net/h : 2,78 bps/h



Interpretation

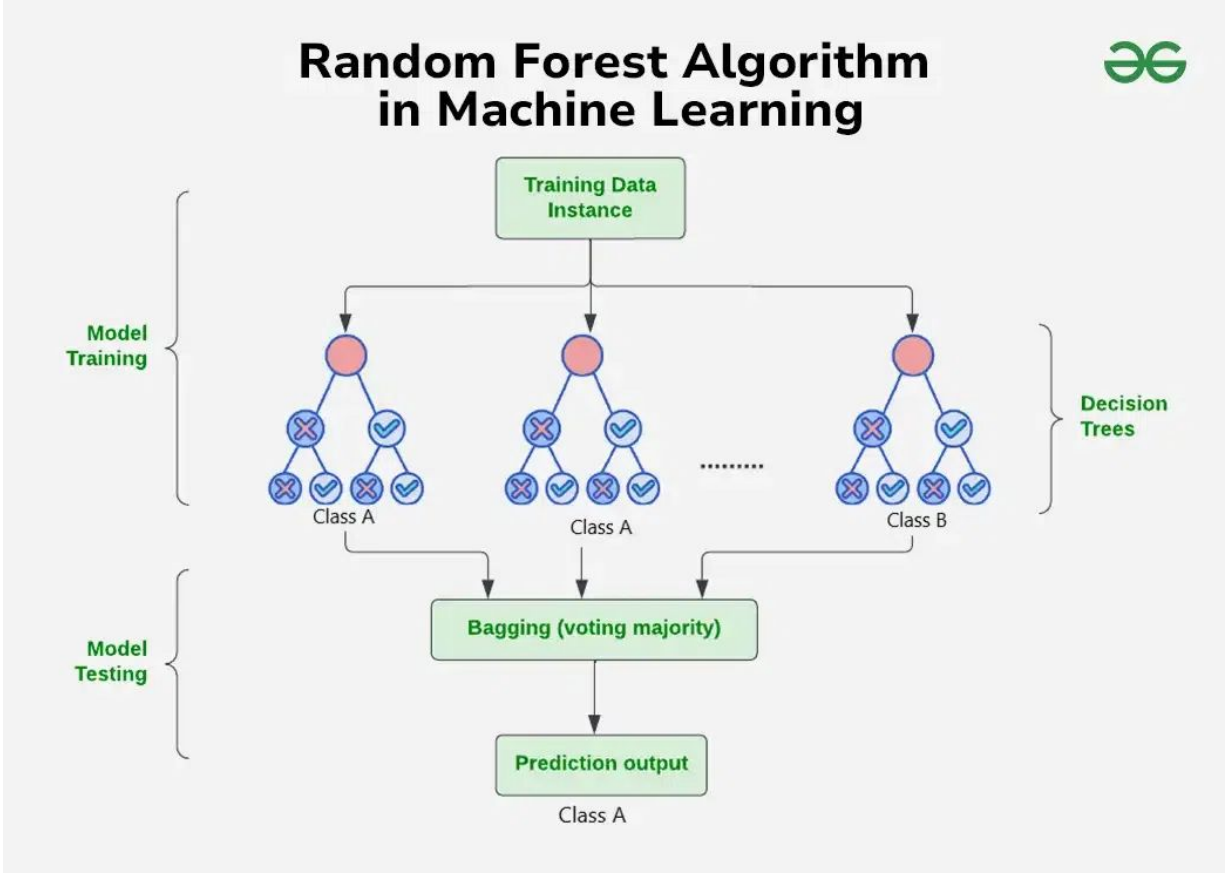
The initial signal :

- Statistically significant TE
- A statistically significant signal is not necessarily exploitable after costs

Future work :

- Maybe the model is too simple (using only 2 features)
- Random forest based strategy

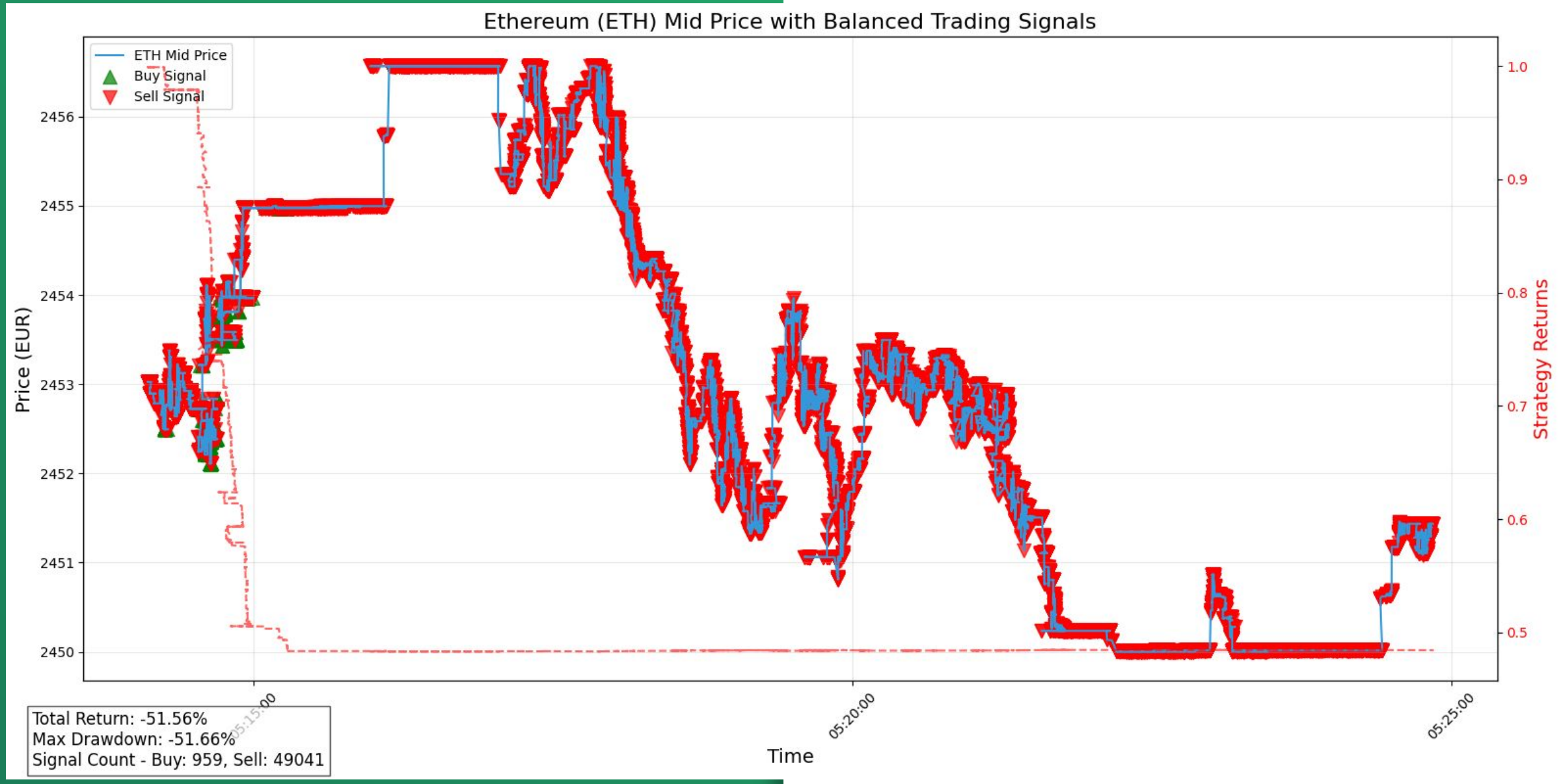
Random Forest based Strategy



Training of the model on 70% of DATASET_0 and testing on the last 30%.



Backtest on DATASET_1 and DATASET_2 to assess performances


Ethereum (ETH) Mid Price with Balanced Trading Signals
ETH Mid Price
Buy Signal
Sell Signal
Price (EUR)
Strategy Returns
Time
2456
2455
2454
2453
2452
2451
2450
1.0
0.9
0.8
0.7
0.6
0.5
05:15:00
05:20:00
05:25:00
Total Return: -51.56%
Max Drawdown: -51.66%
Signal Count - Buy: 959, Sell: 49041

Biased data : The price of ETH is decreasing

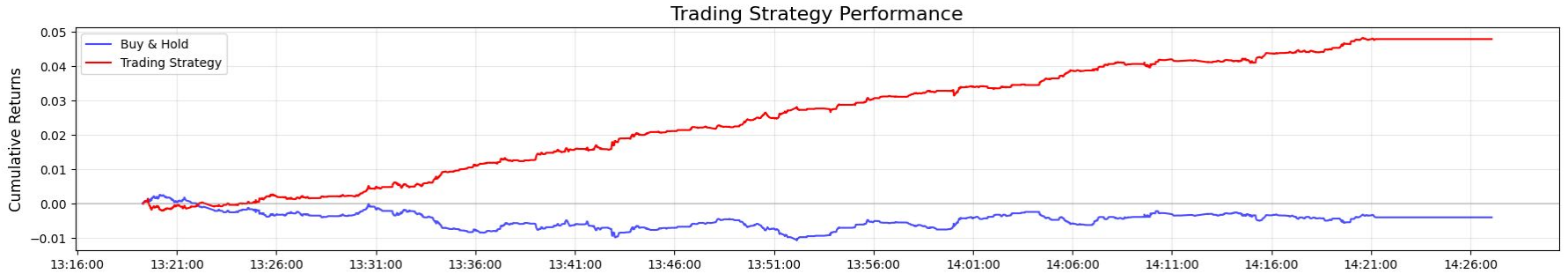
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Class Distribution in Training Set: [10544   546]
Majority Class Ratio: 0.9508
```

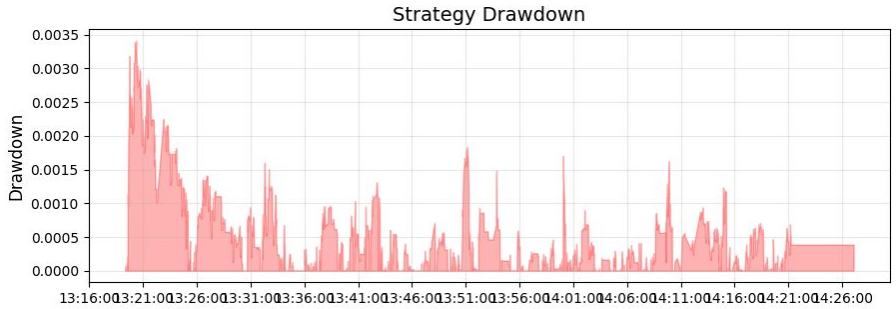


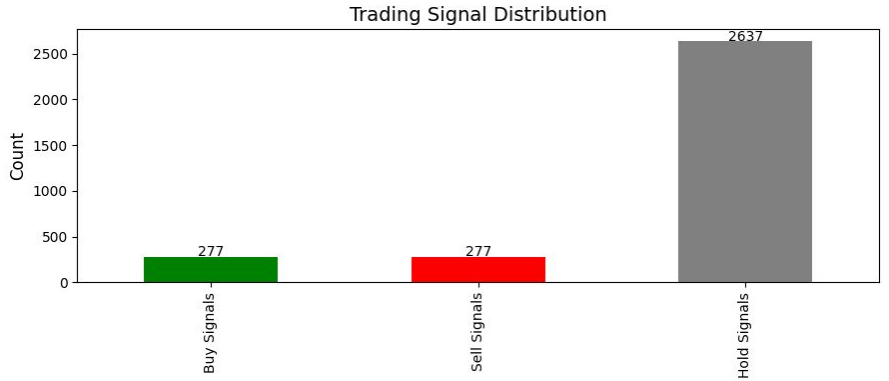
Weighted data included
Training window enlarged



Performance Evaluation on DATASET_1 (Without fees and without latency on transactions):

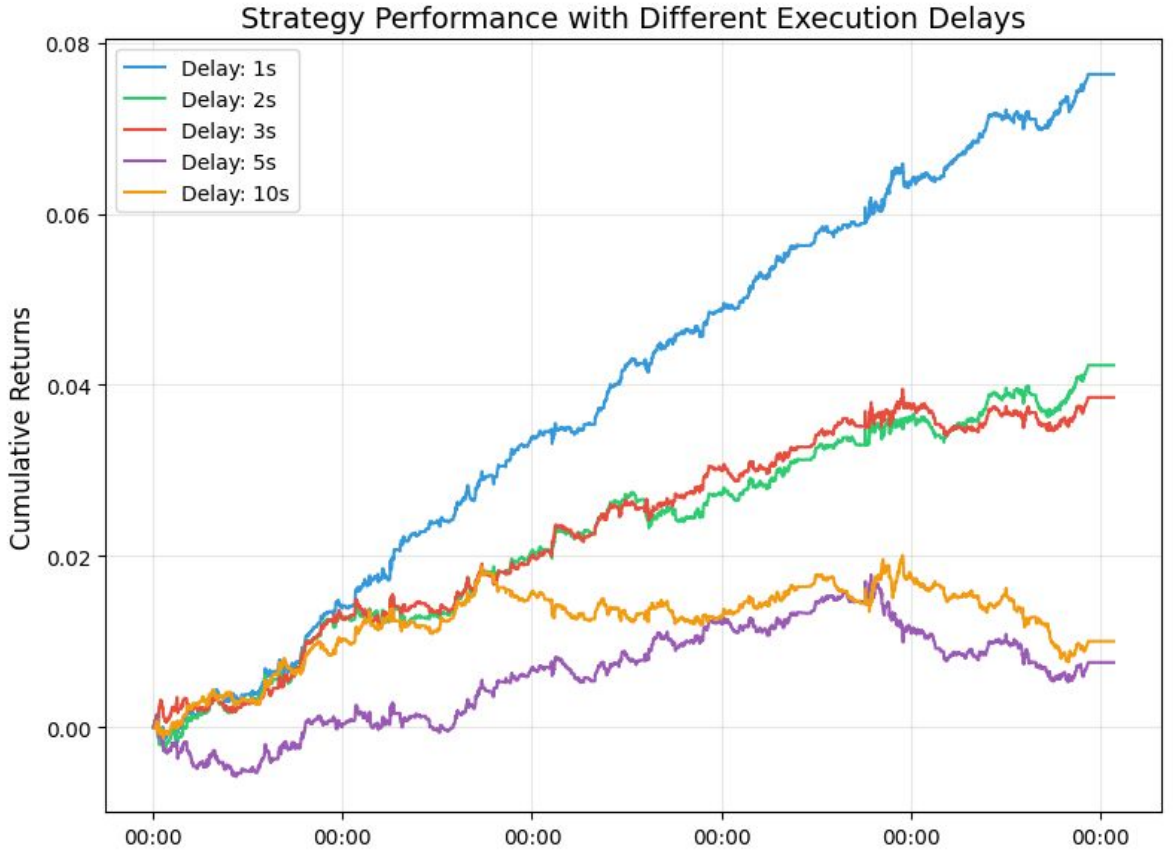






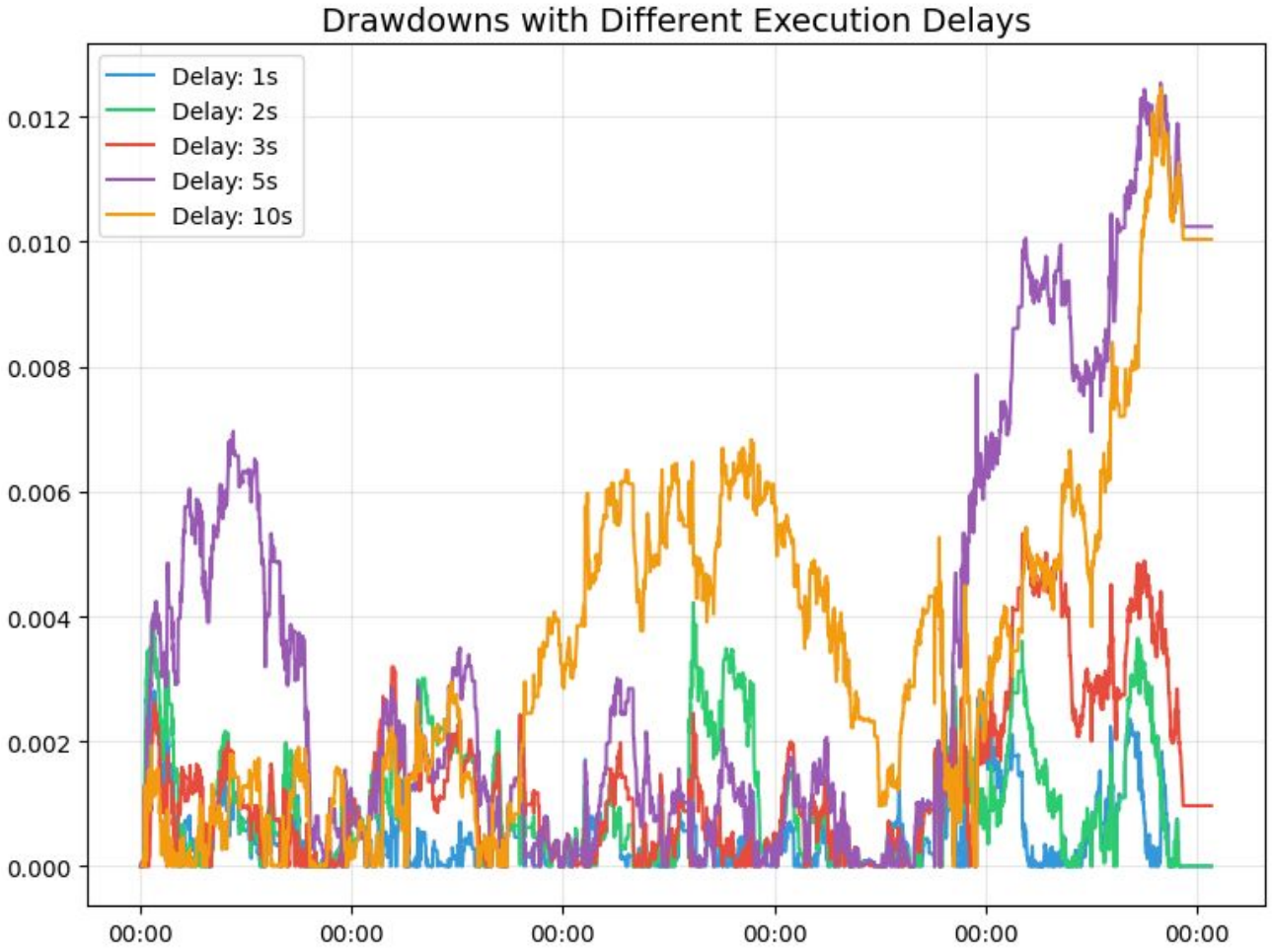


Impact of delay on model without fees (1)

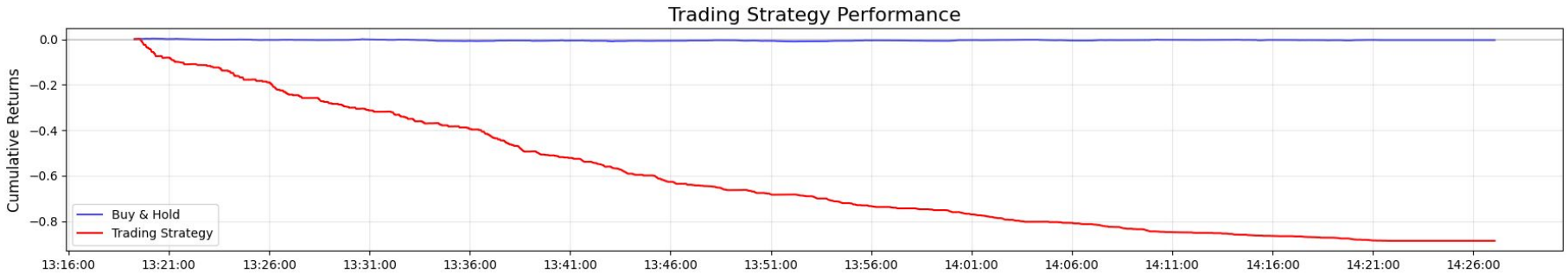


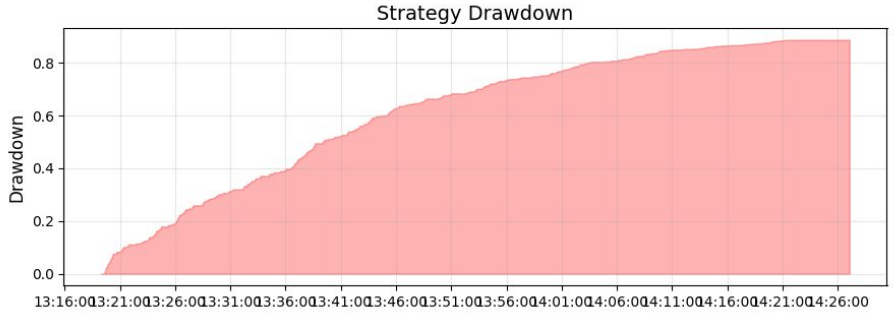
Delay (seconds)	Total Return (%)
1	7.629871
2	4.229777
3	3.854760
5	0.757435
10	1.004375

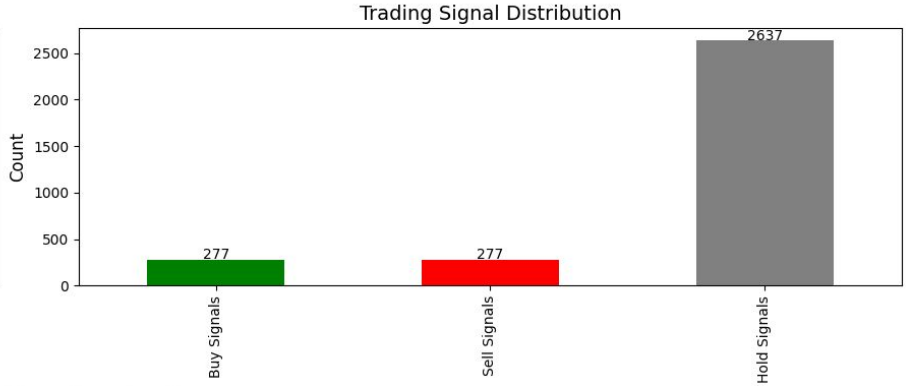
Impact of delay on model without fees (2)



Performance Evaluation on DATASET_1 (With transaction fees and 1 second latency):

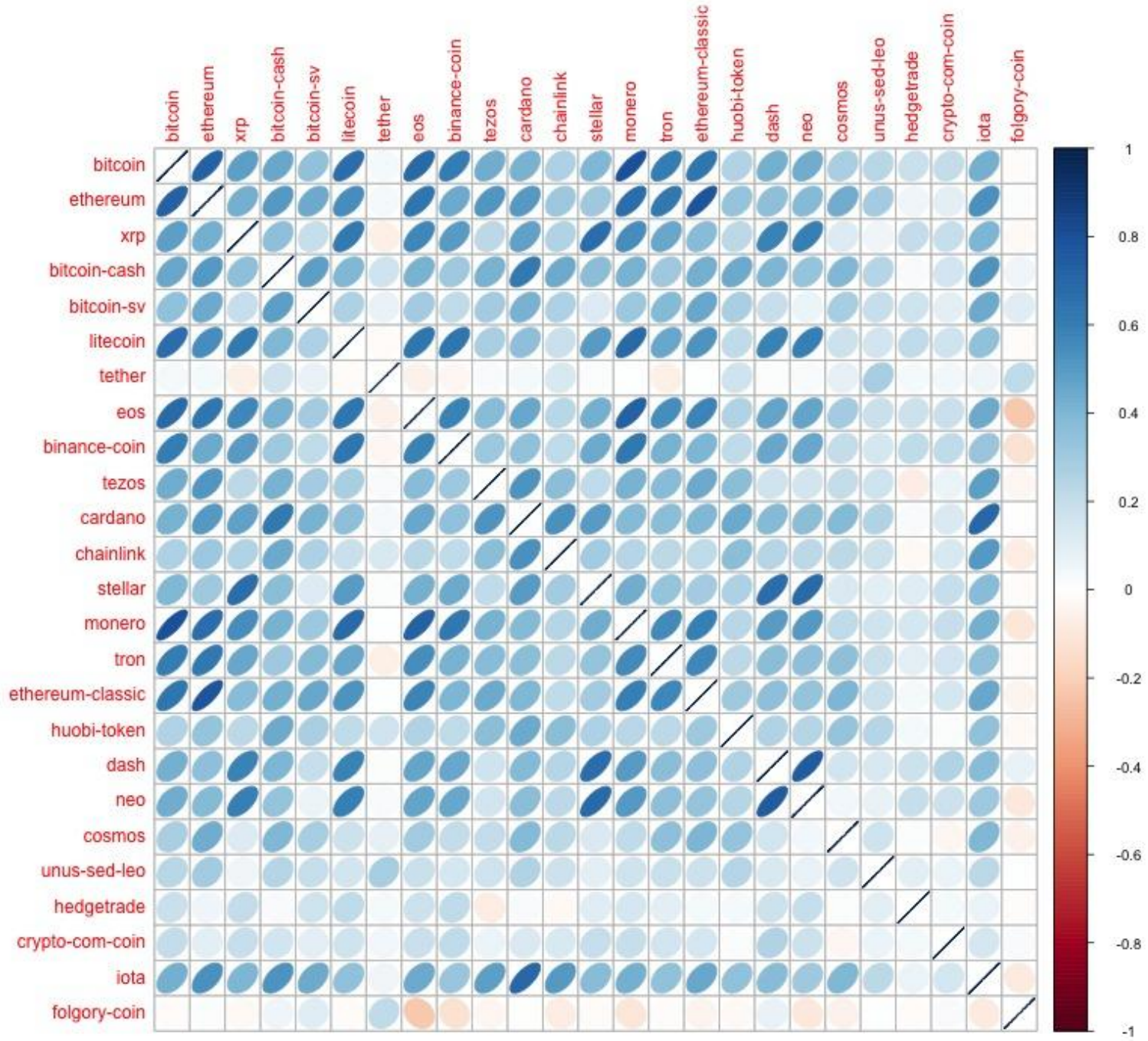








Possible optimization



Considering others coins :

Possible future optimizations :

- Take into consideration real-time delay on transactions (which depends on volumes on market orders)
- Train and test more models based on XGBOOST, LightGBM…



Thanks for listening.

Why does both give different results ?

Transfer Entropy

- Is more concerned with causality than Random Forest (“"Does past X predict future Y better than past Y itself?")
- Gives the direction of the causality

Random Forest

- More about prediction than causality (“If we add X to our model, does it improve predictions of Y”)
- Helps determine how much each feature is contributing to the prediction of Y